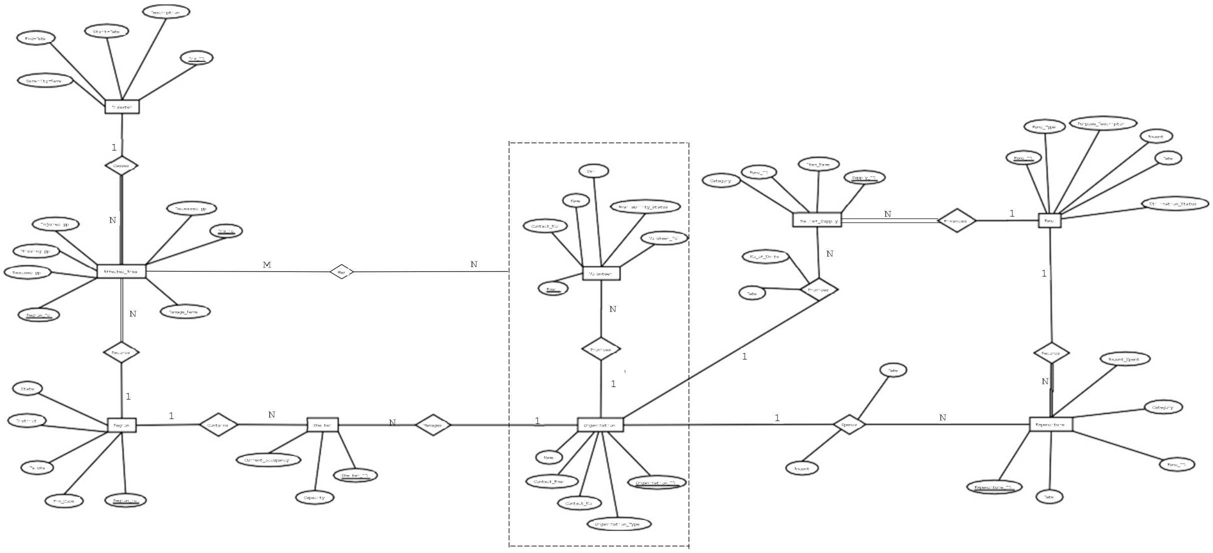
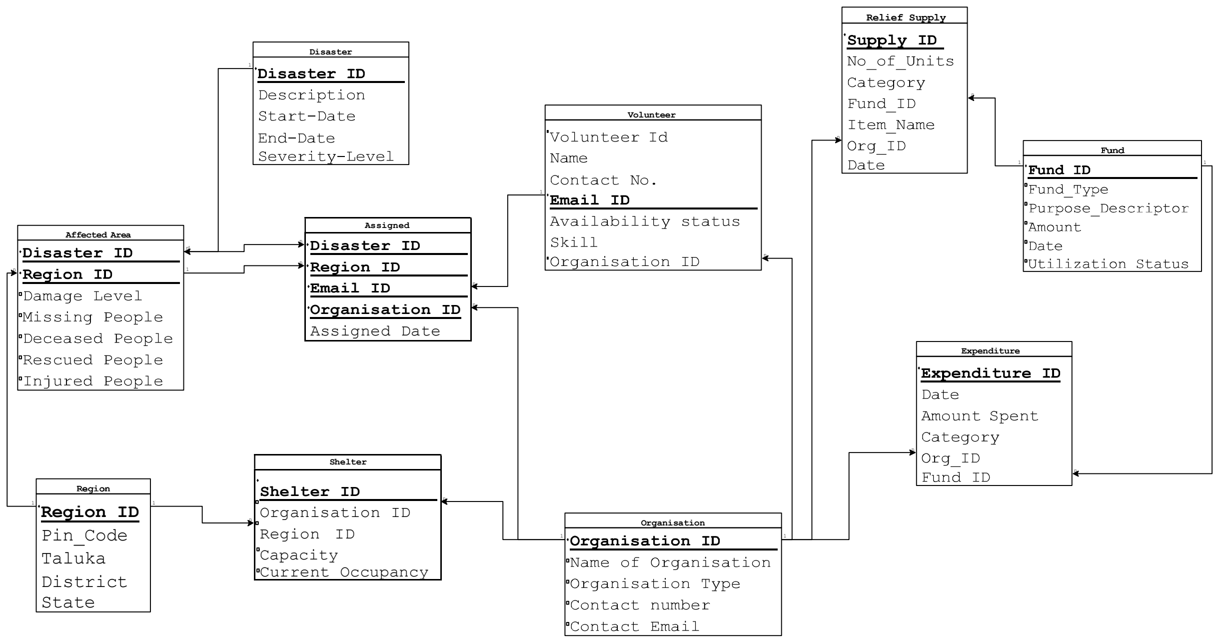
IT 214

Disaster Relief Management System



Start-Date
End-Date
Causes
Affected_Area
Damage_Level
Records
State
Region
Pin_Code
Contains
Current_Occupancy
Capacity
Manages
Has
Name
Contact_No
Contact_Email
Organisation
Organisation_Type
Provides
Category
Item_Name
No_of_Units
Date
Finances
Fund
Fund_Type
Amount
Spends
Expenditure
Amount_Spent
1
N
M

1. Relational Schema



Functional Dependence Analysis

Part – 1 : List of Functional Dependence for Entities

1--- (Disaster)

FDs:

```
Dis_ID → Start_Date, End_Date, Severity_Level, Description, Damage_Level,
Region_ID
```

```
Dis_ID → Deceased_ppl, Injured_ppl, Missing_ppl, Rescued_ppl
```

Notes:

Casualty figures recorded per disaster via the 'Records' relationship.

Region_ID is a FK → Region (1:N — each disaster belongs to one region).

2--- (Region)

FDs:

```
Region_ID → District, Taluka, Pin_Code, State, Affected_Area
```

3--- (Shelter)

FDs:

```
Shelter_ID → Capacity, Current_occupancy
Shelter_ID → Region_ID
```

Notes:

Each shelter is contained within one region (Contains: 1:N)

4--- (Relief_Supply)

FDs:

```
Supply_ID → Category, No_of_Units, Item_Name
Supply_ID → Fund_ID
```

Notes:

Fund_ID is FK → Fund (supply is financed by one fund).

5--- (Fund)

FDs:

```
Fund_ID → Fund_Type, Purpose_Descriptor, Amount, Date, Utilization_Status
Fund_ID → Organization_ID
```

Notes:

Organization_ID is FK → Organization (fund is financed by one organization).

Fund_ID → Name, Organization_Type, Contact_No, Contact_Email *TRANSITIVE via Organization_ID+

6--- (Expenditure)

FDs:

```
Expenditure_ID → Category, Amount_Spent, Date
Expenditure_ID → Fund_ID
```

Notes:

Fund_ID is FK → Fund (expenditure spends from one fund).

Expenditure_ID → Fund_Type, Amount, Utilization_Status *TRANSITIVE via Fund_ID+

7--- (Organization)

FDs:

```
Organization_ID → Name, Organization_Type, Contact_No, Contact_Email
```

8--- (Volunteer)

FDs:

```
Email_ID → Skill, Name, Contact_No, Availability_status
```

Notes:

Volunteer is now on the M side of TWO M:N relationships:

(a) Volunteer — Has — Disaster (M:N) → junction table created

(b) Volunteer — Provides — Organization (M:N) → junction table created

Therefore Volunteer does NOT carry Disaster_ID or Organization_ID as FK.

9--- (NEW junction table – M:N 'HAS' relation)

FDs:
`(Email_ID, Dis_ID) → Date, Amount`

Notes:

Composite PK: (Email_ID, Dis_ID).

Date and Amount are relationship attributes on the 'Has' relationship.

Both columns are also FKs: Email_ID → Volunteer, Dis_ID → Disaster.

10--- (NEW junction table – M:N 'PROVIDES' relation)

FDs:

`(Email_ID, Organization_ID) → [no extra attributes]`

Transitive FDs present in Full Set:

Transitive FDs	Derivation Chain
Dis_ID → District, Taluka, Pin_Code, State, Affected_Area	Dis_ID → Region_ID → Region attributes
Fund_ID → Name, Org_Type, Contact_No, Contact_Email	Fund_ID → Organization_ID → Organization attributes
Expenditure_ID → Fund_Type, Amount, Utilization_Status	Expenditure_ID → Fund_ID → Fund attributes
Supply_ID → Fund_Type, Amount, Utilization_Status	Supply_ID → Fund_ID → Fund attributes

Part – 2 : Minimal FD Set for the Entities

1--- (Disaster)

Minimal FD Set:

Dis_ID → Start_Date
Dis_ID → End_Date
Dis_ID → Severity_Level
Dis_ID → Description
Dis_ID → Damage_Level
Dis_ID → Region_ID
Dis_ID → Deceased_ppl
Dis_ID → Injured_ppl
Dis_ID → Missing_ppl
Dis_ID → Rescued_ppl

2--- (Region)

Minimal FD Set:

Region_ID → District
Region_ID → Taluka
Region_ID → Pin_Code
Region_ID → State
Region_ID → Affected_Area

3--- (Shelter)

Minimal FD Set:

Shelter_ID → Capacity
Shelter_ID → Current_occupancy
Shelter_ID → Region_ID

4--- (Relief_Supply)

Minimal FD Set:

Supply_ID → Category
Supply_ID → No_of_Units
Supply_ID → Item_Name
Supply_ID → Fund_ID

5--- (Fund)

Minimal FD Set:

Fund_ID → Fund_Type
Fund_ID → Purpose_Descriptor
Fund_ID → Amount
Fund_ID → Date
Fund_ID → Utilization_Status
Fund_ID → Organization_ID

6--- (Expenditure)

Minimal FD Set:

Expenditure_ID → Category
Expenditure_ID → Amount_Spent
Expenditure_ID → Date
Expenditure_ID → Fund_ID

7--- (Organization)

Minimal FD Set:

Organization_ID → Name
Organization_ID → Organization_Type
Organization_ID → Contact_No
Organization_ID → Contact_Email

8--- (Volunteer)

Minimal FD Set:
Email_ID → Skill
Email_ID → Name
Email_ID → Contact_No
Email_ID → Availability_status

9--- (Volunteer_Disaster – M:N junction with 'HAS' relation)

Minimal FD Set:

(Email_ID, Dis_ID) → Date
(Email_ID, Dis_ID) → Amount

10--- (Volunteer_Organization – M:N junction with 'PROVDES' relation)

Minimal FD Set:

No non-key attributes — PK (Email_ID, Organization_ID) is the only FD

Part – 3 : FDs removed for the Minimal FD Set

Removed FDs	Type	Available Derivable path
Dis_ID → District, Taluka, Pin_Code, State, Affected_Area	Transitive	Dis_ID → Region_ID → Region attributes
Fund_ID → Name, Organization_Type, Contact_No, Contact_Email	Transitive	Fund_ID → Organization_ID → Org attributes
Expenditure_ID → Fund_Type, Amount, Purpose_Descriptor, Utilization_Status	Transitive	Expenditure_ID → Fund_ID → Fund attributes
Supply_ID → Fund_Type, Amount, Purpose_Descriptor, Utilization_Status	Transitive	Supply_ID → Fund_ID → Fund attributes
Volunteer.Disaster_ID (as FK) Volunteer.Organization_ID (as FK)	Structural Change	M:N relationships replaced with junction tables; FKs moved to junction tables

---

BCNF Proof For Each Relation

1--- (Disaster)

Schema:
Disaster(Disaster_ID, Description, Start_Date, End_Date, Severity_Level, Region_ID)

FDs:
Disaster_ID → Description, Start_Date, End_Date, Severity_Level, Region_ID

BCNF Verification:

Only FD present: Disaster_ID → {all other attributes}

Disaster_ID is the Primary Key, hence a Superkey

Every determinant (Disaster_ID) is a Superkey

No partial dependency — single-attribute PK

No transitive dependency — Region_ID is a FK, not a determinant here

Hence, Relation is in BCNF.

2--- (Region)

Schema:

`Region(Region_ID, Pin_Code, Taluka, District, State)`

FDs:

`Region_ID → Pin_Code, Taluka, District, State`

BCNF Verification:

Only FD present: `Region_ID → {Pin_Code, Taluka, District, State}`

`Region_ID` is the Primary Key, hence a Superkey

Every determinant is a Superkey

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

3--- (Shelter)

Schema:

`Shelter(Shelter_ID, Capacity, Current_Occupancy, Region_ID)`

FDs:

`Shelter_ID → Capacity, Current_Occupancy, Region_ID`

BCNF Verification:

Only FD present: `Shelter_ID → {Capacity, Current_Occupancy, Region_ID}`

`Shelter_ID` is the Primary Key, hence a Superkey

`Region_ID` is a FK referencing Region — it does not determine any attribute here

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

4--- (Organization)

Schema:

```
Organization(Organization_ID, Name, Contact_No, Contact_Email,
Organization_Type)
```

FDs:

```
Organization_ID → Name, Contact_No, Contact_Email, Organization_Type
```

BCNF Verification:

Only FD present: `Organization_ID → {all attributes}`

`Organization_ID` is the Primary Key, hence a Superkey

No non-prime attribute determines another non-prime attribute

Hence, Relation is in BCNF.

5--- (Volunteer)

Schema:

```
Volunteer(Volunteer_ID, Name, Skill, Contact_No, Email,
Availability_Status, Disaster_ID, Organization_ID)
```

FDs:

```
Email_ID → Name, Skill, Contact_No, Availability_Status, Disaster_ID,
Organization_ID
```

BCNF Verification:

Only FD present: `Email_ID → {all other attributes}`

`Email_ID` is the Primary Key, hence a Superkey

`Disaster_ID` and `Organization_ID` are FKs — they do not determine any attribute

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

6--- (Fund)

Schema:

```
Fund(Fund_ID, Fund_Type, Purpose_Descriptor, Amount, Date,
Utilization_Status)
```

FDs:

```
Fund_ID → Fund_Type, Purpose_Descriptor, Amount, Date, Utilization_Status
```

BCNF Verification:

Only FD present: `Fund_ID → {all attributes}`

`Fund_ID` is the Primary Key, hence a Superkey

No non-key attribute determines another attribute

Hence, Relation is in BCNF.

7--- (Relief_Supply)

Schema:

```
Relief_Supply(Supply_ID, No_of_Units, Category, Item_Name, Fund_ID, Org_ID)
```

FDs:

```
Supply_ID → No_of_Units, Category, Item_Name, Fund_ID, Org_ID
```

BCNF Verification:

Only FD present: `Supply_ID → {all other attributes}`

`Supply_ID` is the Primary Key, hence a Superkey

`Fund_ID` and `Org_ID` are FKs — they do not determine any other attribute here

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

8--- (Affected_Area)

Schema:

```
Affected_Area(Disaster_ID, Deceased_Ppl, Injured_Ppl, Missing_Ppl,
Rescued_Ppl)
```

FDs:

```
Disaster_ID → Deceased_Ppl, Injured_Ppl, Missing_Ppl, Rescued_Ppl
```

BCNF Verification:

`Disaster_ID` is the Primary Key (FK referencing Disaster), hence a Superkey

Casualty attributes depend solely on `Disaster_ID`

No non-key attribute determines another

Hence, Relation is in BCNF.

DDL Script for the below Relational Schema

```
-- DISASTER

CREATE TABLE Disaster (
    Disaster_ID INT PRIMARY KEY,
    Description VARCHAR(255),
    Start_Date DATE,
    End_Date DATE,
    Severity_Level VARCHAR(50)
);


-- REGION


CREATE TABLE Region (
    Region_ID INT PRIMARY KEY,
    Pin_Code VARCHAR(10),
    Taluka VARCHAR(100),
    District VARCHAR(100),
    State VARCHAR(100)
);
```

```sql
-- ORGANISATION

CREATE TABLE Organisation (
    Organisation_ID INT PRIMARY KEY,
    Name VARCHAR(255),
    Organisation_Type VARCHAR(100),
    Contact_Number VARCHAR(15),
    Contact_Email VARCHAR(100)
);


-- FUND

CREATE TABLE Fund (
    Fund_ID INT PRIMARY KEY,
    Fund_Type VARCHAR(100),
    Purpose_Descriptor VARCHAR(255),
    Amount DECIMAL(12,2),
    Date DATE,
    Utilization_Status VARCHAR(50)
);

-- RELIEF_SUPPLY

CREATE TABLE Relief_Supply (
    Supply_ID INT PRIMARY KEY,
    No_of_Units INT,
    Category VARCHAR(100),
```

```
    Item_Name VARCHAR(255),
    Date DATE,
    Fund_ID INT,
    Org_ID INT,
    FOREIGN KEY (Fund_ID) REFERENCES Fund(Fund_ID),
    FOREIGN KEY (Org_ID) REFERENCES Organisation(Organisation_ID)
);

-- EXPENDITURE

CREATE TABLE Expenditure (
    Expenditure_ID INT PRIMARY KEY,
    Date DATE,
    Amount_Spent DECIMAL(12,2),
    Category VARCHAR(100),
    Org_ID INT,
    Fund_ID INT,
    FOREIGN KEY (Org_ID) REFERENCES Organisation(Organisation_ID),
    FOREIGN KEY (Fund_ID) REFERENCES Fund(Fund_ID)
);

-- VOLUNTEER

CREATE TABLE Volunteer (
    Volunteer_ID INT ,
    Name VARCHAR(255),
    Contact_No VARCHAR(15),
    Email_ID VARCHAR(100) PRIMARY KEY,
    Availability_Status VARCHAR(50),
```

```
    Skill VARCHAR(100),
    Organisation_ID INT,
    FOREIGN KEY (Organisation_ID) REFERENCES Organisation(Organisation_ID)
);

-- SHELTER

CREATE TABLE Shelter (
    Shelter_ID INT PRIMARY KEY,
    Organisation_ID INT,
    Region_ID INT,
    Capacity INT,
    Current_Occupancy INT,
    FOREIGN KEY (Organisation_ID) REFERENCES Organisation(Organisation_ID),
    FOREIGN KEY (Region_ID) REFERENCES Region(Region_ID)
);

-- AFFECTED_AREA

CREATE TABLE Affected_Area (
    Disaster_ID INT,
    Region_ID INT,
    Damage_Level VARCHAR(50),
    Missing_People INT,
    Deceased_People INT,
    Rescued_People INT,
    Injured_People INT,
    PRIMARY KEY (Disaster_ID, Region_ID),
    FOREIGN KEY (Disaster_ID) REFERENCES Disaster(Disaster_ID),
```

```sql
    FOREIGN KEY (Region_ID) REFERENCES Region(Region_ID)
);

-- ASSIGNED (Relationship table)

CREATE TABLE Assigned (
    Disaster_ID INT,
    Region_ID INT,
    Organisation_ID INT,
    Email_ID VARCHAR(100),
    Assigned_Date DATE,
    PRIMARY KEY (Disaster_ID, Region_ID, Organisation_ID),
    FOREIGN KEY (Disaster_ID) REFERENCES Disaster(Disaster_ID),
    FOREIGN KEY (Region_ID) REFERENCES Region(Region_ID),
    FOREIGN KEY (Organisation_ID) REFERENCES Organisation(Organisation_ID),
    FOREIGN KEY (Email_ID) REFERENCES Volunteer(Email_ID)
);
```